



SHARIF UNIVERSITY OF TECHNOLOGY

Department of Mechanical Engineering

Design, Sizing, and Construction Report of an Industrial Hydraulic Power Pack

Multi-Cylinder Electro-Hydraulic System Synthesis with Modular Sandwich-Plate Manifolds

Course: Hydraulics and Pneumatics

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1. Project Specifications & Circuit Topology

This engineering report presents the comprehensive design, catalog-based component sizing, SolidWorks 3D packaging, and reservoir construction for an industrial electro-hydraulic power unit (HPU). The power unit is designed to independently operate three double-acting hydraulic cylinders under a nominal continuous working pressure of 150 bar at a controlled linear speed of 5 cm/s.

1.1 Initial Hydraulic Schematic (Phase 1 Conception)

The initial circuit topology was conceived using standard ISO 1219 hydraulic symbology. It incorporates three independent actuator branches equipped with dual meter-out flow throttling and pilot-operated check valves for secure load-holding, alongside an unloaded idle circulation line and primary pressure relief.

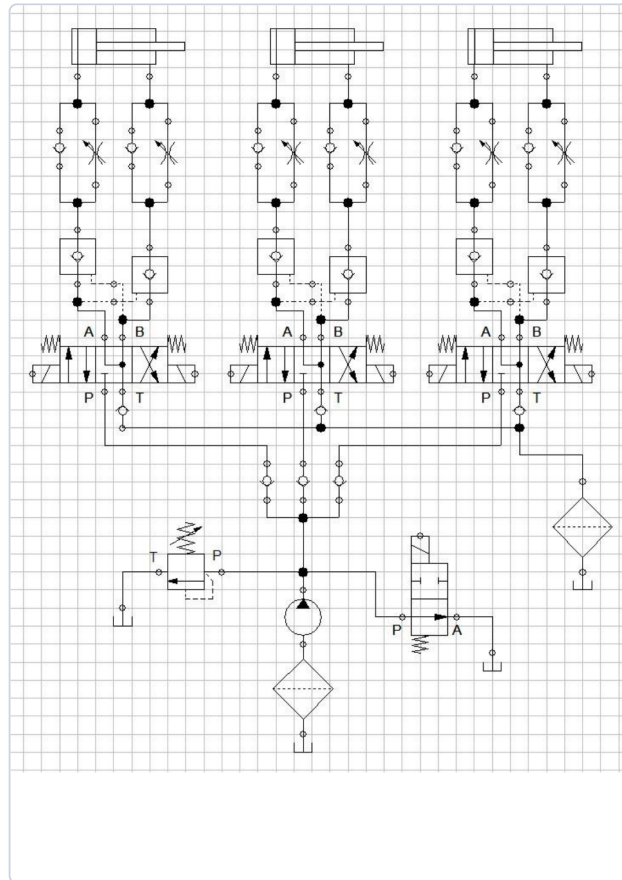


Figure 1: Complete Hydraulic Circuit Diagram across 3 Actuator Lines with Safety and Unloading Branches.

2. Actuator Sizing & Euler Buckling Analysis

The hydraulic cylinders require double-acting operation with a piston bore of 100 mm and a stroke of 1000 mm. Pin-clevis mounting (Type B) was selected to provide pivotal freedom during motion.

2.1 Cylinder Force Analysis:

$$\text{Piston Bore Area: } A_p = \pi \times (100 \text{ mm})^2 / 4 = 78.54 \text{ cm}^2 = 7.854 \times 10^{-3} \text{ m}^2$$

$$\text{Extension Thrust (Push): } F_{push} = P \times A_p = 150 \times 10^5 \text{ Pa} \times 7.854 \times 10^{-3} \text{ m}^2 = 117.81 \text{ kN}$$

From the **Bosch Rexroth CD210 Series** catalog, a piston rod diameter of 45 mm was chosen (rod area = 15.90 cm², annulus area = 62.60 cm², pull retraction force = 93.95 kN at 150 bar).

2.2 Piston Rod Buckling Verification (Euler Theory):

For double-ended pin clevises, the effective column factor is $K = 1.0$. With stroke length $X^* = 1000 \text{ mm}$ and retracted pin offset $XN = 294 \text{ mm}$, maximum extended column span is:

$$L_{max} = X^* + XN = 1000 \text{ mm} + 294 \text{ mm} = 1294 \text{ mm} = 1.294 \text{ m}$$

$$\text{Second Moment of Area: } I = \pi \times D^4 / 64 = \pi \times (45 \text{ mm})^4 / 64 = 2.013 \times 10^5 \text{ mm}^4 = 2.013 \times 10^{-7} \text{ m}^4$$

$$\text{Euler Critical Buckling Load: } P_{cr} = \pi^2 E I / L^2 = [\pi^2 \times (210 \times 10^9 \text{ Pa}) \times (2.013 \times 10^{-7} \text{ m}^4)] / (1.294 \text{ m})^2 \approx 249.0 \text{ kN}$$

Buckling Safety Factor: Since $F_{push} = 117.81 \text{ kN} < P_{cr} = 249.0 \text{ kN}$, the design exhibits an Euler safety factor of **S.F. = 2.11**. The rod is verified to be fully safe against compressive buckling under maximum working pressure.

Type of mounting B Operating pressure 210 bar

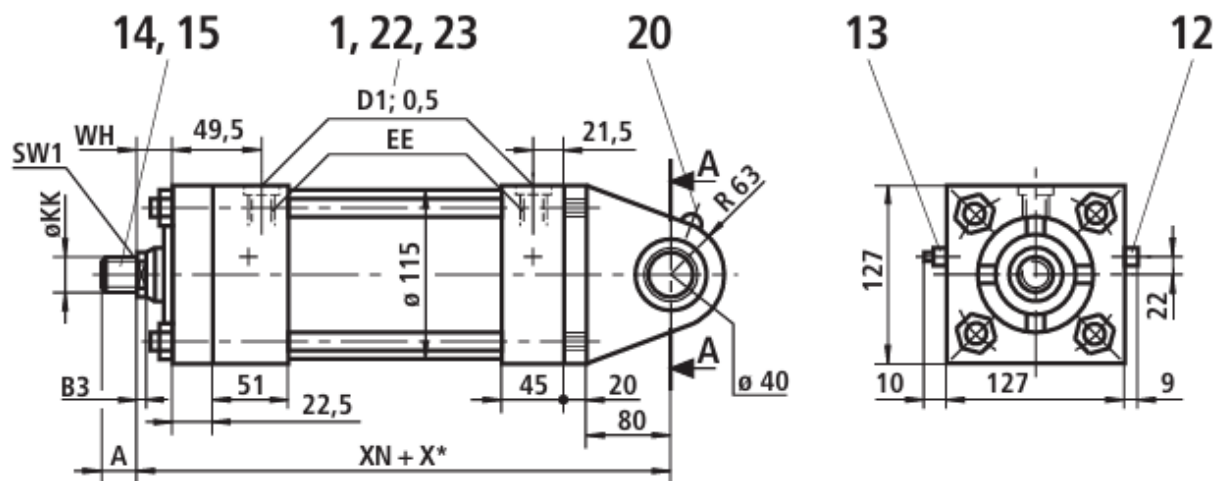


Figure 2a: Rexroth CD210 Clevis Type B Cylinder Dimensions.

3. Hydraulic Pump, Motor & Drive Assembly

3.1 Flow Demand & Pump Sizing

For an actuator linear speed of $v = 5 \text{ cm/s}$ (0.05 m/s), required volumetric flow rate per operating cylinder is:

$$Q = v \times A_p = 5 \text{ cm/s} \times 78.54 \text{ cm}^2 = 392.7 \text{ cm}^3/\text{s} = 23.56 \text{ L/min}$$

From the **Casappa Polaris PLP 20** catalog, pump model **PLP 20•14** was selected, featuring:

- Displacement: $V_g = 14.53 \text{ cm}^3/\text{rev}$ (Max speed: 3500 RPM, Continuous pressure: 250 bar > 150 bar target).
- Speed required to supply 23.56 L/min: $n = 23,560 / 14.53 = 1623 \text{ RPM}$.
- Shaft power required at 150 bar (from Casappa characteristic curves at 46 cSt): $P_{\text{pump}} \approx 6.5 \text{ kW}$.
- Mounting: **SAE "A" 2-Bolt** flange with straight keyed shaft $\varnothing 19 \text{ mm}$ (Key $4.76 \times 28.5 \text{ mm}$).

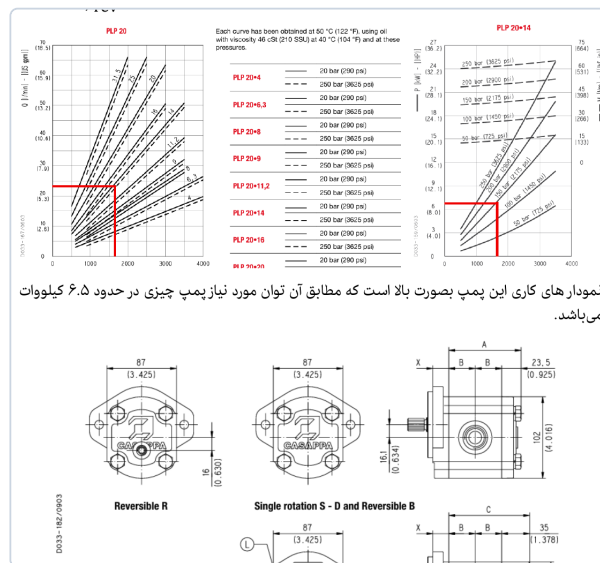


Figure 3: Casappa Polaris PLP 20 Characteristic Delivery and Shaft Input Power Curves.

3.2 Electric Motor & Flexible Coupling Selection

With mechanical transmission efficiency $\eta_t = 0.85$, motor power is $P_{\text{motor}} = 6.5 / 0.85 = 7.64 \text{ kW} \approx 7.5 \text{ kW}$. The chosen prime mover is the **Motogen Tabriz 132M4B** 3-phase induction motor:

- Output Power: **7.5 kW (10 HP)**, 380V Δ / 660V Y, 4-Pole, 50 Hz, Rated Speed: 1445 RPM.
- Full-load torque: $T_{\text{nom}} = 49.56 \text{ N}\cdot\text{m}$. Applying a service factor of 1.5 yields $T_{\text{design}} \approx 75 \text{ N}\cdot\text{m}$.
- Coupling: **KTR ROTEX® 28 Type 1b** flexible jaw coupling (92 ShA spider rated at $95 \text{ N}\cdot\text{m} > 75 \text{ N}\cdot\text{m}$, length 140 mm accommodating shaft diameters of $\varnothing 38 \text{ mm}$ and $\varnothing 19 \text{ mm}$).

3.3 Custom Bellhousing Design

To interface the IEC 132M B5 motor flange ($\varnothing 300 \text{ mm}$, $\varnothing 265 \text{ mm}$ PCD) with the SAE "A" 2-bolt pump flange ($\varnothing 82.55 \text{ mm}$ spigot, $\varnothing 106.2 \text{ mm}$ bolt span) while enclosing the 140 mm coupling, a custom rigid bellhousing was modeled and detailed in SolidWorks.

Technical drawing of a Bellhousing (1000000) showing front, side, and top views with dimensions.

Front View (Top):

- Outer diameter: $\varnothing 150.00$
- Inner diameter: $\varnothing 110.00$
- Central hole diameter: $\varnothing 42.33$
- Inner hole diameter: $\varnothing 20.00$
- Distance from center to outer edge: 130.00
- Distance from center to inner hole: 130.00

Side View (Right):

- Overall width: 140.00
- Height: 100.00
- Distance from center to outer edge: 12.00

Top View (Bottom):

- Outer diameter: $\varnothing 150.00$
- Inner diameter: $\varnothing 110.00$
- Distance from center to outer edge: 140.00
- Distance from center to inner hole: 130.00

Table 1: Dimensions and Tolerances

Item	Dimension	Tolerance
1	140.00	± 0.10
2	100.00	± 0.10
3	12.00	± 0.10
4	130.00	± 0.10
5	130.00	± 0.10
6	140.00	± 0.10
7	130.00	± 0.10
8	130.00	± 0.10

Table 2: Material and Finish

Item	Material	Finish
1	Aluminum	Matte
2	Aluminum	Matte
3	Aluminum	Matte
4	Aluminum	Matte
5	Aluminum	Matte
6	Aluminum	Matte
7	Aluminum	Matte
8	Aluminum	Matte

Table 3: Assembly and Drawing Information

Item	Assembly	Drawing
1	Assembly	Drawing
2	Assembly	Drawing
3	Assembly	Drawing
4	Assembly	Drawing
5	Assembly	Drawing
6	Assembly	Drawing
7	Assembly	Drawing
8	Assembly	Drawing

Table 4: Revision History

Rev	Description	Date
1	Initial release	10/10/2010
2	Revision 1	10/10/2010
3	Revision 2	10/10/2010
4	Revision 3	10/10/2010
5	Revision 4	10/10/2010
6	Revision 5	10/10/2010
7	Revision 6	10/10/2010
8	Revision 7	10/10/2010

Table 5: Drawing Information

Item	Description	Value
1	Part Name	Bellhousing
2	Part Number	1000000
3	Part Description	Aluminum
4	Part Material	Aluminum
5	Part Finish	Matte
6	Part Weight	1.00
7	Part Volume	1.00
8	Part Length	1.00
9	Part Width	1.00
10	Part Height	1.00

Table 6: Assembly and Drawing Information

Item	Assembly	Drawing
1	Assembly	Drawing
2	Assembly	Drawing
3	Assembly	Drawing
4	Assembly	Drawing
5	Assembly	Drawing
6	Assembly	Drawing
7	Assembly	Drawing
8	Assembly	Drawing

Table 7: Revision History

Rev	Description	Date
1	Initial release	10/10/2010
2	Revision 1	10/10/2010
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5	Revision 4	10/10/2010
6	Revision 5	10/10/2010
7	Revision 6	10/10/2010
8	Revision 7	10/10/2010

Table 8: Drawing Information

Item	Description	Value
1	Part Name	Bellhousing
2	Part Number	1000000
3	Part Description	Aluminum
4	Part Material	Aluminum
5	Part Finish	Matte
6	Part Weight	1.00
7	Part Volume	1.00
8	Part Length	1.00
9	Part Width	1.00
10	Part Height	1.00

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	2/2 PUPPET VALVE	M-2SED6PK1X/350CG24K4	1
2	RELIEF VALVE	ZDB6DP3-2X/200	1
3	MANIFOLD	1HSR04-2S/01CFE//2N8//CN/T0	1

Exploded view diagram of a 4/3S valve assembly. The diagram shows the main valve body (1) with various ports and connections. Callout 1 points to the top of the valve body. Callout 2 points to the top of the valve body. Callout 3 points to the top of the valve body. Callout 4 points to the top of the valve body. Callout 5 points to the bottom of the valve body.

ITEM NO.	PART NUMBER	DESCRIPTION
1	4/3S POOL VALVE	4WE6/JAX/EO
2	PLOT OPERATED CHECK VALVE	1256-1-A
3	METER-OUT	2ZF56-3-4X
4	CHECK VALVE	Z15AP0-4
5	MANIFOLD	1HSR06 25/01C/P2/NE

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5. Power Pack Layout & Sump Construction in SolidWorks

The complete hydraulic power pack assembly was fully modeled and packaged in SolidWorks. The reservoir holds a nominal volume of **120 Liters**, exceeding the standard industrial guideline of 3 to 5 times the pump flow rate (3 times 23.56 approx 71 ext{ L}). External top-cover dimensions are 700 mm $\times$ 500 mm with 10 mm plate thickness.

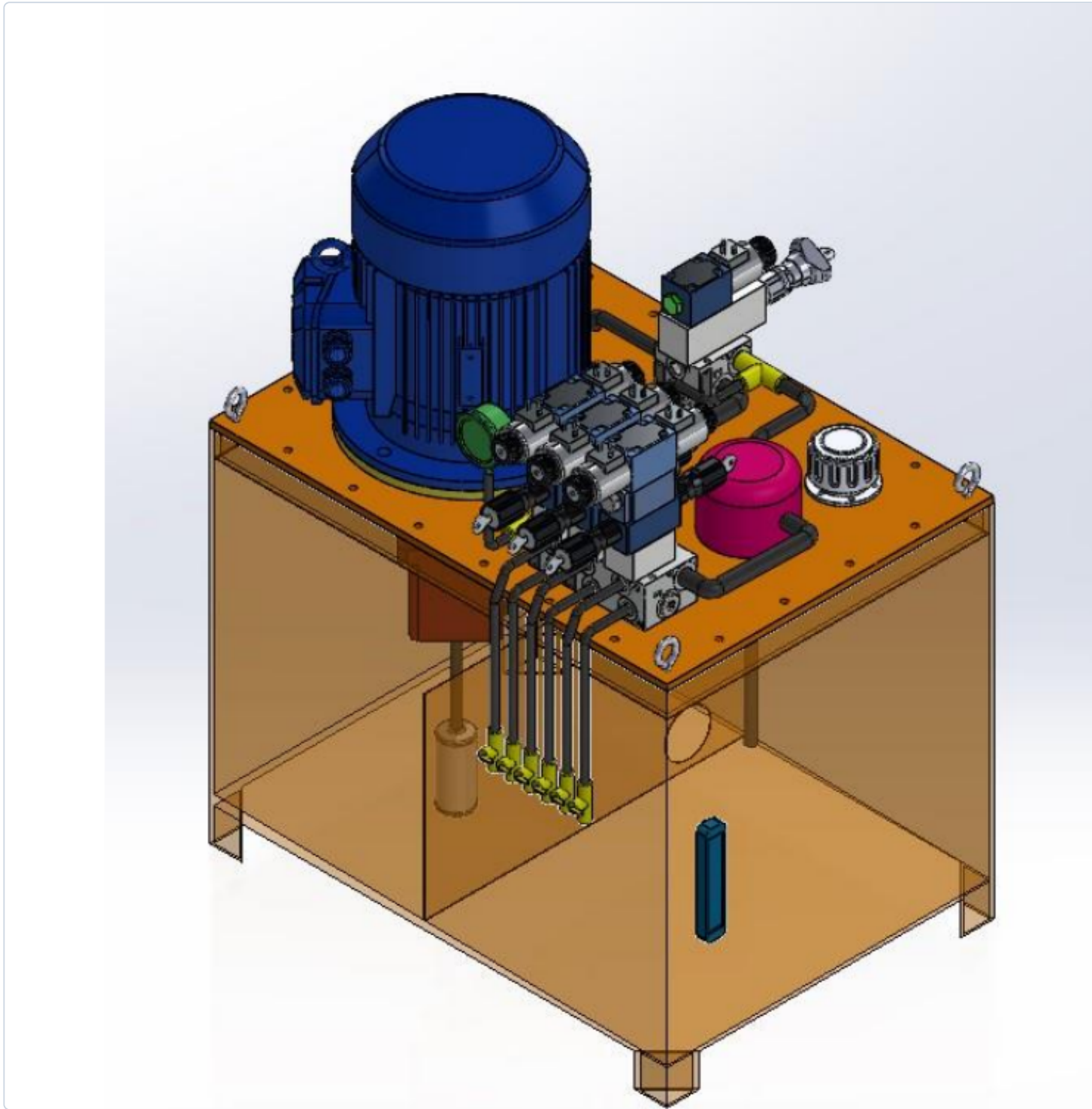


Figure 6a: SolidWorks 3D Isometric View of Hydraulic Power Pack Assembly.

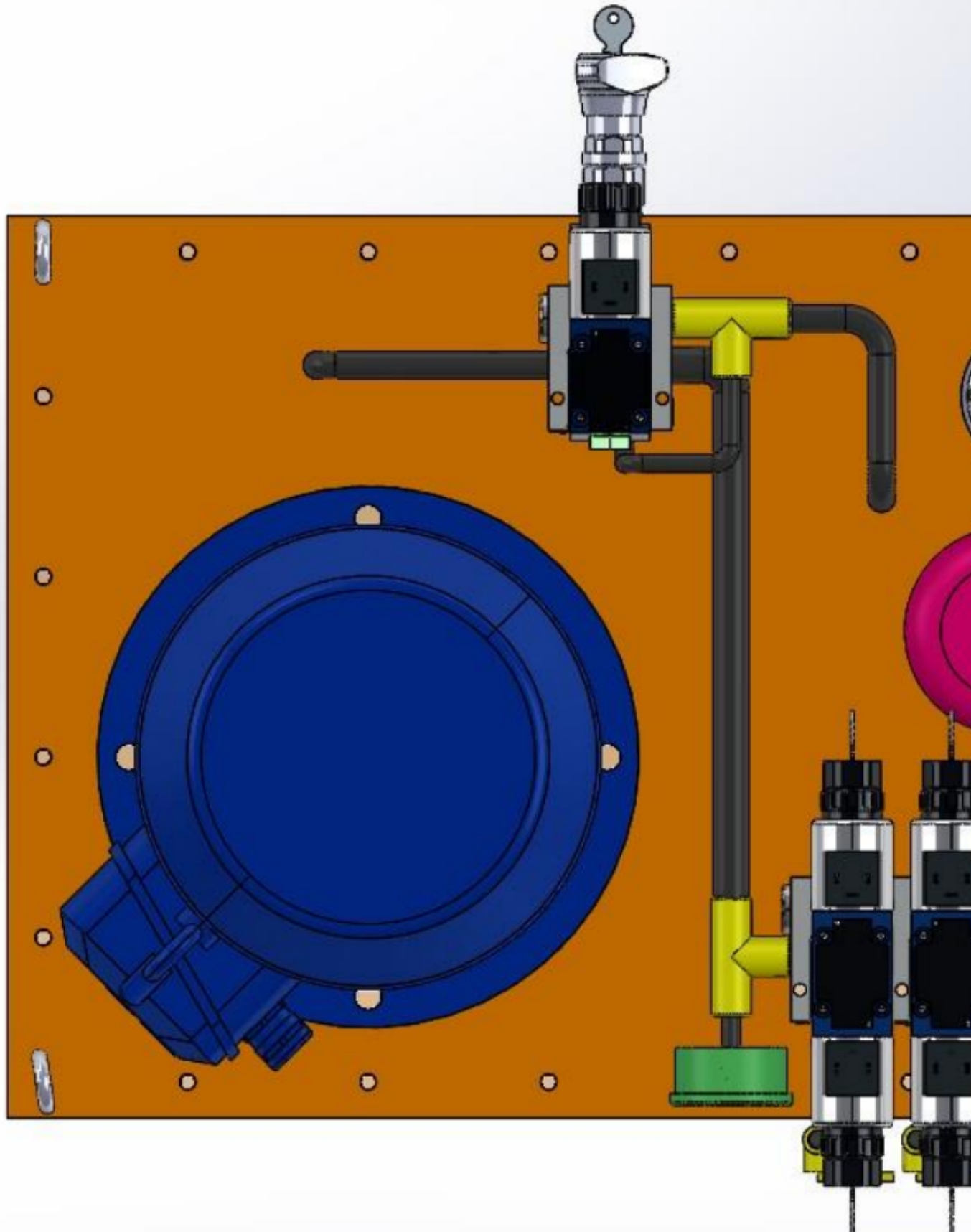


Figure 6c: Top-Down View showing Component Layout on Sump Cover.

5.1 Critical Sump Construction Guidelines

- **Internal Baffle & Rat-Holes:** A vertical baffle plate isolates the suction zone from hot return fluid. Notched lower corners (rat-holes) allow smooth equalization during draining without stagnant dead-zones.
- **Sloped Sump Bottom:** The reservoir bottom slopes toward a lowest collection point with a magnetic drain plug to trap metallic debris, sludge, and condensation water.
- **Line Submergence:** The return pipe extends at least 20 mm below the lowest operating oil level (cut at 45°) to eliminate oil aeration and foaming. The suction strainer is immersed while positioned >20 mm above the floor to avoid sucking settled sludge.
- **Elevation & Lifting:** Sturdy 50 mm feet elevate the tank for underfloor convective cooling. Four M12 DIN 580 eyebolts on the lid enable balanced overhead crane rigging during maintenance.

6. Interconnect Piping & Bill of Materials (BOM)

All hard piping utilizes Ø12 mm seamless precision steel tubing with heavy-duty DIN 2353 / ISO 8434-1 bite-type fittings adhering to British Standard Pipe (BSP Parallel / ISO 1179-2) threads.

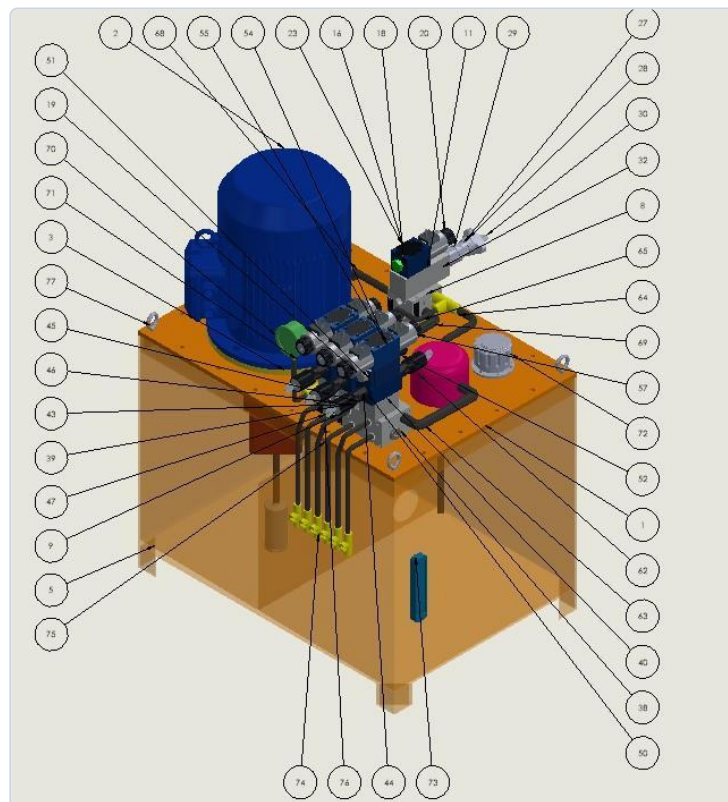


Figure 7: Exploded Itemized CAD Callout Diagram of Hydraulic Power Pack Assembly.

Item	Component Description	Manufacturer Part / Model No.	Manufacturer	Qty	Technical Specifications
1	Hydraulic Cylinder	CD210 Series (Type B Clevis)	Bosch Rexroth	3	Ø100/45 mm, Stroke 1000 mm, 210 bar
2	Hydraulic Gear Pump	PLP 20•14 (SAE "A" Flange)	Casappa	1	14.53 cc/rev, 250 bar, Ø19 mm shaft

Item	Component Description	Manufacturer Part / Model No.	Manufacturer	Qty	Technical Specifications
3	Electric Induction Motor	132M4B (B5 Flange FF265)	Motogen Tabriz	1	7.5 kW, 4-Pole, 1445 RPM, 380/660V
4	Flexible Jaw Coupling	ROTEX® 28 Type 1b (92 ShA)	KTR	1	T _{nom} 95 Nm, L=140 mm, bores Ø38/Ø19
5	Bellhousing (Pump Adapter)	Custom SolidWorks Design (A3)	In-House / Custom	1	IEC 132M B5 to SAE "A", H=140 mm
6	3-Station NG6 Manifold	3HSR06-25/01CFE//ZN8//CN/TO	Bosch Rexroth	1	P, T = G1/2; A, B = G3/8; 315 bar
7	1-Station NG6 Manifold	1HSR06-25/01CFE//ZN8//CN/TO	Bosch Rexroth	1	P, T = G1/2; A, B = G3/8; 315 bar
8	4/3 Directional Spool Valve	4WE6J6X/EG24K4	Bosch Rexroth	3	NG6, Spool J, 24V DC Solenoids
9	Pilot Operated Double Check	Z2S6-1-6X	Bosch Rexroth	3	NG6 Sandwich, 1.5 bar cracking
10	Meter-Out Throttle Check	Z2FS6-3-4X/20	Bosch Rexroth	3	NG6 Sandwich, calibrated rotary knob
11	Sandwich Check Valve	Z1S6P05-4X/V	Bosch Rexroth	3	NG6 Sandwich, Channel P, 0.5 bar
12	2/2 Solenoid Poppet Valve	M-2SED6PK1X/350CG24K4	Bosch Rexroth	1	NG6 Poppet, 350 bar, 24V DC
13	System Pressure Relief Valve	ZDB6DP3-2X/200	Bosch Rexroth	1	NG6 Sandwich, P-T, 200 bar, knob
14	Suction Strainer	STRC04510M0090AN00FG01200NNNP01	MP Filtri	1	90 µm wire mesh, G1/2 connection
15	Return Line Filter	10TEN0040-H6XLA00-P2.2-M-R3	Bosch Rexroth	1	6 µm glass fiber, G3/4, mech indicator
16	Pressure Gauge (Glycerin)	ABZMM63-160BAR/MPA-U/V-G	Bosch Rexroth	1	Ø63 mm, 0–160 bar, bottom G1/4 port
17	Breather / Filler Cap	AB116310	Parker Hannifin	1	10 µm filtration, 7.5 L/s air flow, 3" dia
18	Fluid Level & Temp Gauge	Visual Level Gauge w/ Thermometer	Standard Ind.	1	Side-mount, Min/Max level marks
19	Straight Male Connector	DMC 12L-03G	Parker Hannifin	7	Ø12 mm Tube × G3/8 Male, 315 bar

Item	Component Description	Manufacturer Part / Model No.	Manufacturer	Qty	Technical Specifications
20	Straight Male Connector	DMC 12L-04G	Parker Hannifin	4	Ø12 mm Tube × G1/2 Male, 315 bar
21	Straight Male Connector	DMC 12L-06G	Parker Hannifin	2	Ø12 mm Tube × G3/4 Male, 315 bar
22	Swivel Run Tee	DEL 12L	Parker Hannifin	2	Ø12 mm Tube Swivel Tee, 315 bar
23	Adjustable 90° Elbow	DLA 12L	Parker Hannifin	1	Ø12 mm Standpipe Elbow, 315 bar
24	Female Gauge Connector	DFC 10L-02G	Parker Hannifin	1	Ø10/12 mm Tube × G1/4 Female
25	Threaded Bulkhead Nipples	Nipple 12 mm / G1/2	Standard Ind.	4	Through-cover bulkhead sealing
26	Isolating Ball Valves	1/4-Turn Ball Valve Ø12 mm	Standard Ind.	6	Actuator service line isolation
27	Lifting Eye Bolts	Forged Steel Eye Bolts M12	DIN 580	4	Balanced top-cover crane rigging
28	Sump Tank Body & Lid	Welded Carbon Steel (700×500×550 mm)	Fabricated	1	Vol ≈ 120 L, sloped floor, baffle plate

7. Conclusion

The electro-hydraulic power unit was systematically designed and validated. By shifting to Bosch Rexroth NG6 sandwich-plate modular manifold blocks, an exceptionally robust, vibration-resistant, and leakage-minimized hydraulic circuit was achieved. The calculations confirm a push force of 117.8 kN with an Euler buckling safety factor of 2.11, pump flow of 23.56 L/min powered by a 7.5 kW motor, and complete thermal and filtration management within a 120 L reservoir.